Matching

CS1200: Introduction to Algorithms and their Limitations
Lecture 14

Fall 2026

Harvard SEAS

Put away your devices

Announcements

- Last 10 minutes today: Problem Set 4 in-class follow-up.
- Reading: Hesterberg–Vadhan Chapter 15 (today); Chapter 16 for Wednesday.
- Wednesday 10/28: Embedded EthiCS — what today's model leaves out.

Recap: when does greedy work?

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Def. 14.1. An independent set in G is a set of vertices with no edge between any two of them.

problem	greedy	best known
Interval Scheduling (§14.3)	optimal, sorted by end time	$O(n \log n)$
Independent Set (§14.1)	can be far from optimal	roughly 1.2^n
Graph 2-Coloring (§13.2)	optimal, in BFS order	$O(n + m)$
Graph 3-Coloring (§14.2)	no guarantee	roughly 1.33^n
Maximum Matching (today)	fails...	...but can be repaired

The problem

Kidney exchange

- US kidney waiting list: about 90,000 patients; a little over 27,000 transplants a year; average wait about 3.6 years.
- A donor gives one kidney, and only to a *compatible* patient (blood type, HLA type).

Model. A compatibility graph: a vertex for each donor and each patient, an edge for each compatible donor–patient pair.

- Before: edges meant *conflict* (registers, intervals). Now an edge means *match*.
- Constraints: each donor gives to at most one patient; each patient receives at most one kidney.
- What kind of graph is it? Bipartite: donors on one side, patients on the other.

Goal: as many transplants as possible.

Definition 15.1: matching

Def. 15.1. A matching in $G = (V, E)$ is a set $M \subseteq E$ such that every vertex is incident to at most one edge of M. Equivalently: no two edges of M share an endpoint.

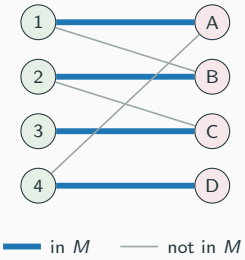
v is matched by M if some edge of M contains it; otherwise unmatched.

Maximum Matching

$\mathcal{I}$: graphs $G = (V, E)$
$\mathcal{O}$: sets of edges M
$f(G) = \{M : M \text{ a matching in } G \text{ of maximum size}\}$

At right: $|M| = 4$. Could any matching be bigger?
No: each edge matches 2 vertices, so $|M| \leq n/2 = 4$.

These definitions are for *any* graph G, not just bipartite.



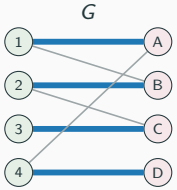
Matching is independent set in disguise (§15.4)

Line graph $L(G)$: one vertex per *edge* of G; two are adjacent if the edges share an endpoint.

Matchings in G = independent sets in $L(G)$.

So Maximum Matching reduces to Independent Set.
Is that a good way to solve it?
No: the best known algorithm for Independent Set is roughly 1.2^n.

Like interval scheduling: an *easy special case* of a hard problem.



Greedy, and how to repair it

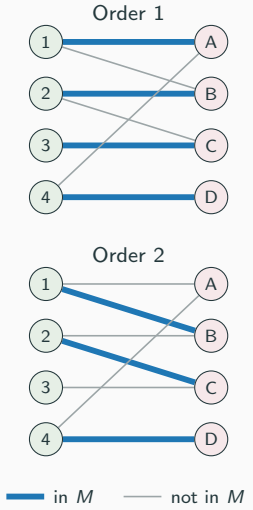
Greedy fails

Greedy: start with $M = \emptyset$; go through the edges in some order; add an edge if it shares no endpoint with M.

Order 1:
$\{1, A\}, \{1, B\}, \{2, B\}, \{2, C\}, \{3, C\}, \{4, D\}, \{4, A\}$.
$|M| = 4$: maximum.

Order 2: $\{4, D\}, \{2, C\}, \{1, B\}$, then the rest.
Every remaining edge touches a matched vertex: maximal (can't add an edge), not maximum.

What went wrong? Greedy never takes an edge back out.
Repair: allow swaps. But which swaps, so that we always make progress?



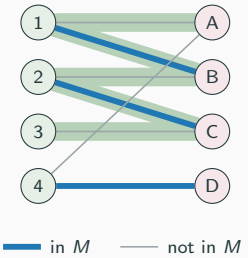
Definition 15.2: alternating walks and augmenting paths

Let M be a matching in G.

An alternating walk (w.r.t. M) is a walk $v_0, v_1, \ldots, v_\ell$ whose edges alternate between in M and not in M.

An augmenting path is an alternating walk with $\ell \geq 1$ such that

- v_0 and v_ℓ are unmatched by M,
- all of its vertices are distinct.



At right: $3, C, 2, B, 1, A$.
Alternating? Yes: out, in, out, in, out.
Augmenting? Yes: 3 and A are unmatched; no repeats.

Any other augmenting paths?
No: only 3 and A are unmatched, and the path between them is forced.

Flipping an augmenting path (Lemma 15.7)

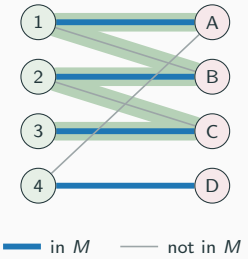
Flip the path: its edges in M leave M; its edges not in M join M.

Lemma 15.7. If P is an augmenting path w.r.t. M, flipping P gives a matching M' with $|M'| = |M| + 1$, in time $O(n)$.

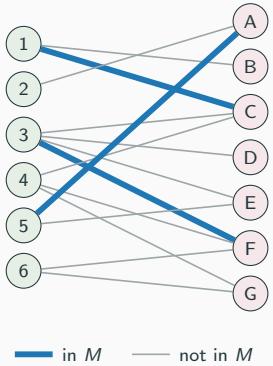
Why $+1$? P starts and ends with non-M edges: one more out than in.

Why still a matching? Inner vertices trade one M-edge for one; the ends were unmatched.

Here: size 3 $\rightarrow$ size 4, the maximum.



In groups: alternating? augmenting?



$M = \{\{1, C\}, \{3, F\}, \{5, A\}\}$

For each sequence: is it an alternating walk? An augmenting path?

sequence	alternating walk?	augmenting path?
$2, A, 5$	yes	no: 5 is matched
$G, 4$	yes	yes
$3, F, 4, C, 3$	no: $\{F, 4\}, \{4, C\} \notin M$	no
$1, C, 6$	no: $\{C, 6\} \notin E$	no
$6, F, 3, C, 1, B$	yes	yes

Berge's Theorem

Berge's Theorem (Theorem 15.3)

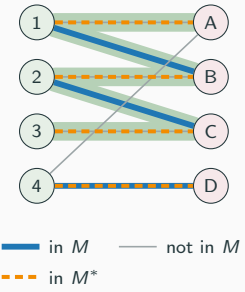
Theorem 15.3 (Berge). A matching M in a graph G is not of maximum size if and only if G has an augmenting path w.r.t. M.

$\Leftarrow$**:** Lemma 15.7 (flip it).

$\Rightarrow$**, proof idea.** Take a bigger matching M^*. Keep only the edges in exactly one of M, M^*: $M \triangle M^*$.

- Every vertex has degree ≤ 2 in it: at most one edge from each matching.
- So it splits into paths and cycles, alternating M / M^*.
- Cycles have as many M edges as M^* edges. $|M^*| > |M|$, so some path has more M^* edges: it starts and ends with M^* edges — an augmenting path for M.

Holds for *every* graph, not just bipartite ones.



$\{4, D\}$ is in both, so it drops out.

The algorithm

The algorithm (Algorithm 15.1)

MaxMatchingAugPaths(G)

Input: a bipartite graph G

0 remove isolated vertices from G
1 let V_0, V_1 be the two sides (a 2-coloring)
2 $M = \emptyset$
3 **repeat**
4 $U_0, U_1 =$ vertices of V_0, V_1 unmatched by M
5 find an augmenting path P from U_0 to U_1 (or $\perp$)
6 **if** $P \neq \perp$: flip P (Lemma 15.7)
7 **until** $P = \perp$
8 **return** M

Correct: it stops only when there is no augmenting path, so by Berge, M is maximum.

Left to do: line 5. How do we *find* an augmenting path?

Finding an augmenting path: shortest paths (Lemma 15.6)

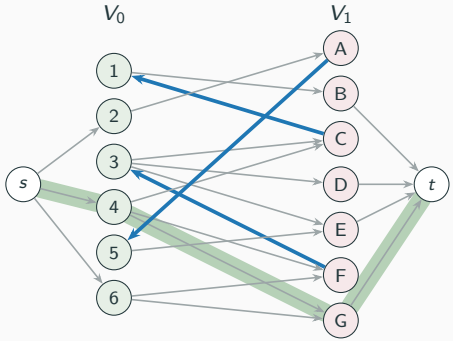
Build a *directed* graph G':

1. edges not in M: point $V_0 \to V_1$;
2. edges in M: point $V_1 \to V_0$;
3. new s, with $s \to u$ for each $u \in U_0$;
4. new t, with $u \to t$ for each $u \in U_1$.

Why does an $s \to t$ path alternate? Lines 1 and 2 above.

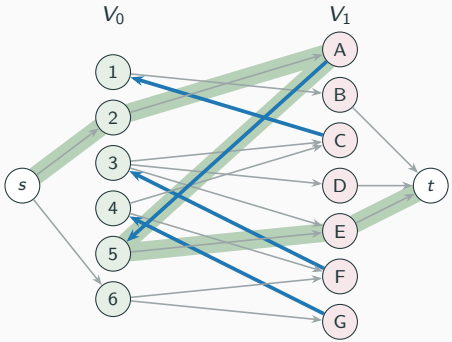
BFS from s: $s \to 4 \to G \to t$.
Drop s, t: augmenting path $(4, G)$ — one edge, a greedy step.

What does the algorithm do on the first round if the matching is empty?
Line 2 starts at $M = \emptyset$. It mimics greedy in the early rounds.



$M = \{\{1, C\}, \{3, F\}, \{5, A\}\}$; $U_0 = \{2, 4, 6\}$, $U_1 = \{B, D, E, G\}$

In groups: the next augmenting path



$M = \{\{1, C\}, \{3, F\}, \{4, G\}, \{5, A\}\}$

1. What changed in G' since the last slide?
   $G \to 4$ now; $s \to 4$ and $G \to t$ are gone.
2. BFS from s. Frontiers:
   $F_1 = \{2, 6\}$, $F_2 = \{A, F, G\}$,
   $F_3 = \{5, 3, 4\}$, $F_4 = \{C, D, E\}$
   Distance to t: 5
3. A shortest path, and the new M:
   $2, A, 5, E$
   $M = \{\{1, C\}, \{2, A\}, \{3, F\}, \{4, G\}, \{5, E\}\}$

Runtime (Theorem 15.8)

Theorem 15.8. Maximum Matching on bipartite graphs can be solved in time $O(mn)$.

one round: build G' and run BFS	$O(n+m)$
flip the path	$O(n)$
number of rounds	$\leq n/2$ (each adds an edge; $\lvert M\rvert \leq n/2$)
total	$O(n(n+m))$

No isolated vertices (line 0) $\Rightarrow n \leq 2m \Rightarrow O(n(n+m)) = O(mn)$.

Not bipartite? Berge still holds, but we're relying on bipartiteness to find an augmenting path. Efficient algorithms exist (Edmonds' "blossoms"), with similar runtime and much more work.

Takeaway

- Greedy gets stuck at a *maximal* matching. The repair is a smarter move: flip an augmenting path, $+1$ each time.
- Berge: no augmenting path $\iff$ maximum. Proved without finding one.
- Finding one in a bipartite graph is a shortest-path problem: orient the edges, add s and t, run BFS. Total $O(mn)$.
- An easy special case of Independent Set, like interval scheduling.

Wednesday 10/28: Embedded EthiCS — what "maximize the number of transplants" leaves out (Ch 16).

Do now: Problem Set 4 in-class follow-up.

Appendix: proofs

Appendix: proof of Berge's Theorem

$\Leftarrow$. If P is augmenting, flipping it gives a matching of size $|M| + 1$ (Lemma 15.7), so M is not maximum.

$\Rightarrow$. Let M^* be a matching with $|M^*| > |M|$, and let $G_\triangle = (V, M \triangle M^*)$.

1. *Degree* ≤ 2. Each vertex is in at most one edge of M and one of M^*. So every component of $G_\triangle$ is a path or a cycle.
2. *Alternation.* Two edges of $G_\triangle$ at a common vertex cannot both be in M, nor both in M^*; so each path and cycle alternates M / M^*, and each cycle has equally many edges of each.
3. *Counting.* $|M^* \setminus M| > |M \setminus M^*|$, and these edges are split among the components. So some path P has more M^* edges than M edges; since it alternates, its first and last edges are in $M^* \setminus M$.
4. *P is augmenting for M.* Its edges alternate in M / not in M, its vertices are distinct, and each endpoint v is unmatched by M: if $\{v, w\} \in M$, then $\{v, w\} \notin M^*$ (v already has its M^* edge on P), so $\{v, w\} \in G_\triangle$ and v would have degree 2. $\square$

Appendix: proof of Lemma 15.7 (flipping)

Lemma 15.7. Let $P = (v_0, \ldots, v_\ell)$ be augmenting w.r.t. M. Then

$$M' = \big(M \setminus \{\{v_1, v_2\}, \{v_3, v_4\}, \ldots, \{v_{\ell-2}, v_{\ell-1}\}\}\big) \cup \{\{v_0, v_1\}, \{v_2, v_3\}, \ldots, \{v_{\ell-1}, v_\ell\}\}$$

is a matching with $|M'| = |M| + 1$.

Proof. v_0 is unmatched, so $\{v_0, v_1\} \notin M$; likewise $\{v_{\ell-1}, v_\ell\} \notin M$. By alternation, ℓ is odd and the edges of P are out, in, out, $\ldots$, out.

- Vertices not on P: their edges in M and M' are the same.
- v_0, v_ℓ: unmatched before; now in exactly one new edge (distinct vertices, so P uses them once).
- $v_1, \ldots, v_{\ell-1}$: each lost its one M edge (on P) and is in exactly one new edge.

So M' is a matching, and it matches exactly v_0, v_ℓ more: $|M'| = |M| + 1$. □

Appendix: proof of Lemma 15.5

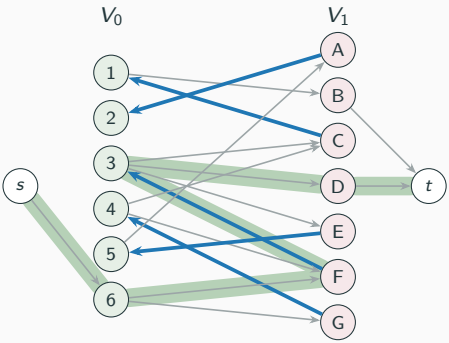
Lemma 15.5. Let G be bipartite with sides V_0, V_1, and M not maximum. Then (1) G has an alternating walk from U_0 to U_1, and (2) every shortest alternating walk from U_0 to U_1 is an augmenting path.

(1). Berge gives an augmenting path $v_0, \ldots, v_\ell$. Its first and last edges are not in M, so ℓ is odd. In a bipartite graph a walk switches sides at each step, so v_0, v_ℓ are on opposite sides: P or its reverse runs from U_0 to U_1.

(2). Let $W = v_0, \ldots, v_\ell$ be shortest, and suppose $v_i = v_j$ with $i < j$. Bipartite: $j - i$ is even, so the loop $v_i, \ldots, v_j$ begins and ends with edges of opposite kinds (M / not M). Cut out $v_{i+1}, \ldots, v_j$: still alternating, and shorter. Contradiction. □

Remark. BFS on G' returns a path, so (2) is automatic there; (1) is what guarantees t is reachable.

Appendix: finishing the run



$M = \{\{1, C\}, \{2, A\}, \{3, F\}, \{4, G\}, \{5, E\}\}$ (after the activity).
$U_0 = \{6\}$, $U_1 = \{B, D\}$.

BFS: $s \to 6 \to F \to 3 \to D \to t$. Augmenting path $6, F, 3, D$.

Flip: $M =$
$\{\{1, C\}, \{2, A\}, \{3, D\}, \{4, G\}, \{5, E\}, \{6, F\}\}$, size 6.

Maximum: every left vertex is matched, and $|M| \leq |V_0| = 6$.
(The next BFS finds no s–t path: $U_0 = \emptyset$.)